Analysis of Covariance

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Analysis of Covariance (ANCOVA)

- With randomization, characteristics of participants and their environment should be balanced across study arms
- However, these characteristics can still affect study outcomes
- Options:
  - Pre-adjust • Incorporate these characteristics into your study design by blocking (blocked randomization)
    - "Stratification" in randomized trial
    - participants are stratified/blocked in those groups, then randomize within those blocks.
    - Ex) stratify into Grad and Undergrad Blocks.
  - Post-adjust • Gather data on these characteristics and in your data analysis, adjust your outcome, Y, for the "covariate", X (ANCOVA)
    - Ex): Extra info on student Status (Grad/Undergrad)
    - Use this extra info to adjust for Y after analysis.

Analysis of Covariance (ANCOVA)

Covariates are often continuous

Linear model with categorical + continuous covariates

Ex) $BP = \mu + Group + Age + \varepsilon$

→ Adjusts group means as if all groups had the same age.

Linear model with categorical predictors only

- Combines regression and ANOVA

Ex) $BP = \mu + Group + \varepsilon$

only compares group means

- Without adjustment, the effects of a covariate X can inflate σ^2
  - Reduce the noise: adjustment can increase precision
  - This could sometimes affect treatment comparisons as well

when you compare treatment groups but know there's a continous "nuisance" factor (covariate) affecting your response, ANCOVA adjusts for it — so treatment comparisons are fairer + more powerful.

ANOVA Pros:

- ↑ Power
- ↓ Error σ^2
- ↓ Confounding
- Adjust Baseline Differences

Analysis of Covariance

Hao Zhang

Introduction

In order to see the difference of treatment effects, we must control the nuisance factors if they are expected to be a major source of variation. If the values of the nuisance factors can be measured in advance of the experiment or controlled during the experiment, then they can be taken into account by using blocking factors. There are situations when some nuisance factors or variables may affect the response and they are also measured. For example, we would like to compare the effects of four diets on the weights of month-old piglets. The response variable is the weight three months after starting the diets, which is likely to be affected by the weight at the beginning of the experiment. We would use the starting weight as a covariate. Therefore the employment of covariates is another way to deal with nuance factors or variables. The analysis of covariance of the study to compare treatment effects in the presence of covariances. Note in terms of design in the example, it is a completely randomized design. What is new here we have additional measurements on the experimental units (piglets' starting weights).

Covariates are often continuous variables. Their effects on the response variable are assumed to follow some parametric form such as linear. → Ex) Depend on x, x^2 ... etc

Model

Model with One Covariate:

For a completely randomized design with one covariate, the simplest model is

$$Y_{ij} = \mu + \underset{\text{Treatment factor}}{\tau_i} + \overset{\text{Linear Covariate}}{\boxed{\beta x_{ij}}} + \epsilon_{ij} \qquad (1)$$

Ex) Linear Quadratic...etc

where Y_{ij} is the response variable of the jth unit from the ith treatment group; τ_i is the effect of the ith treatment, x_{ij} is the value of the covariate of the jth unit in the ith treatment group, ϵ_{ij} are i.i.d. $N\left(0, \sigma^2\right)$.

Here we assume that the effect of the covariate on the response is linear. It can be extended to more complex cases, e.g., quadratic, cubic, or polynomial effects.

$$y_{ij} = \mu + \overset{\text{Treatment factor}}{\tau_i} + \overset{\text{Linear Covariate}}{\underline{\beta x_{ij}}} + \varepsilon_{ij} \begin{cases} i = 1, 2, \ldots, a \\ j = 1, 2, \ldots, n_i \end{cases}$$

Trt group unit

$$\varepsilon_{ij} \sim iid\ N(0, \sigma^2)$$

- Additional assumptions:
  - x_{ij} not affected by the treatments
  - x and y are linearly related
  - For now: constant slopes for each treatment arm
- Sometimes we center the x_{ij}: use $(x_{ij} - \bar{x}_{..})$ instead of x_{ij} to fit the model

- Regular ANOVA has has Trt. and Error. df Error = n-V
- ANCOVA adds extra row (β) for covariate. df Error = n-V-1 (↓ SSE, ↑Power)

Statistical Tests

of groups

Source of variation	Degrees of freedom	Sum of squares	Mean squares	Ratio
T	$v-1$	ssT	$\frac{ssT}{v-1}$	$\frac{msT}{msE}$
β	1	$ss\beta$	$ss\beta$	
Error	$n-v-1$	ssE		
Total	$n-1$	ss_{yy}	msE	

- F tests
- An extra degree of freedom is used by the model to estimate β. This df comes out of df_{error}.

① Source: Trt

- H_0: $\tau_1=\tau_2=\ldots=\tau_a=0$ "No Adjusted Group Difference"
  - Compare treatment means after adjusting for differences among treatments due to differences in covariate levels $\boxed{F=\frac{MS_T}{MS_E}}$

② Source: Covariate

- H_0: $\beta=0$ "Covariate has No Effect"
  - F test will have numerator df=1 and denominator df= df_{error} = $\underline{N-(a+1)}_{N-a-1}$ $\boxed{F=\frac{MS_\beta}{MSE}}$

Mean Estimates

- Adjusted treatment means
  - Based on the expected value of y when x is equal to the average covariate value
  - Can really use any value of x, as long as it's reasonable for all treatments
- Pairwise differences:

differences in mean X

If Trt. groups differ in X, it will correct it.

$$\underbrace{\hat{\tau}_i - \hat{\tau}_{i'}}_{\text{* Adjusted Difference.}} = \overbrace{\underbrace{(\overline{y_{i.}} - \overline{y_{i'.}})}_{\text{originally: only difference of sample means}}}^{\text{Raw Difference}} - \underbrace{\hat{\beta}(\overline{x_{i.}} - \overline{x_{i'.}})}_{\text{New Component: Adjustment for differences in covariate means}}$$

Regression Approach (2-arm CRD)

$$y_j = \beta_o + \beta_1 x_{1j} + \beta_2 x_{2j} + \varepsilon_j$$

$$\varepsilon_j \sim iid\ N(0, \sigma^2)$$

- $j = 1, 2, \ldots, \Sigma n_i$ (full sample, across all groups)
- X_{1j} = 1 if the *j*th observation is in treatment 1

  0 if the *j*th observation is in treatment 2
- X_{2j} = covariate value for the *j*th observation

ANCOVA with Non-constant Slope

- Recall model with constant slope:

• Assume β_2 is the same in either arm.

$$y_{ij} = \mu + \tau_i + \beta x_{ij} + \varepsilon_{ij} \begin{cases} i = 1, 2, \ldots, a \\ j = 1, 2, \ldots, n_i \end{cases}$$

- Allow a different slope for each treatment *i*:

• But you might have a different slope depending on different scenario / trt. arm

Arm-Specific slope

$$y_{ij} = \mu + \tau_i + (\underbrace{\beta}_{\text{generic slope}} + (\beta\tau)_i)x_{ij} + \varepsilon_{ij} \begin{cases} i = 1, 2, \ldots, a \\ j = 1, 2, \ldots, n_i \end{cases}$$

Non-constant Slope:
Regression Approach (2-arm CRD)

$$y_j = \beta_o + \beta_1 x_{1j} + \beta_2 x_{2j} + \beta_3 x_{1j} x_{2j} + \varepsilon_j$$

- $j = 1, 2, \ldots, \Sigma n_i$ (full sample, across all groups)
- X_{1j} = 1 if the *j*th observation is in treatment 1

  0 if the *j*th observation is in treatment 2
- X_{2j} = covariate value for the *j*th observation

Intercept

$Y = \underbrace{(\beta_0 + \beta_1)}_{\text{Intercept}} + \beta_2 X_{2j} + \beta_3 X_{2j} + \varepsilon_j$

$Y = (\beta_0 + \beta_1) + \underline{(\beta_2 + \beta_3) X_{2j}} + \varepsilon_j$

Intercept

$Y = (\beta_0) + \beta_2 X_{2j} + \varepsilon_j$

"ancova.sas"

Ex) : SAS

β_0 { Intercept 26.084

β_1 { Trt 1 -8.175; Trt 2 0.000

β_2 { Baseline 0.568

Constant slope — No interaction — Parallel $\beta_3 = 0$

Intercept different, slope same

Trt 1 — $X_1 = 1$

Intercept

$$Y = (\beta_0 + \underline{\beta_1}) + \beta_2 X_{2j} + \varepsilon_j$$

$$Y = (\beta_0 + \beta_1) + (\beta_2) X_{2j} + \varepsilon_j$$

$$\rightarrow Y = (26.084 \underline{- 8.175}) + 0.568 X_2$$

Trt 2 — $X_2 = 0$

Intercept

$$Y = (\beta_0) + \beta_2 X_{2j} + \varepsilon_j$$

$$\rightarrow Y = 26.084 + 0.568 X_2$$

Different β_0, same β_2 (slope)

Ex) : SAS

β_0 { Intercept 28.198

β_1 { Trt 1 -46.648; Trt 2 0.000

β_2 { Baseline 0.948

β_3 { Baseline × Trt 1 -0.572; Baseline × Trt 2 0.000 — Interaction (β_3)

Nonconstant slope — Interaction (β_3)! $\beta_3 \neq 0$

Intercept different, slope different

Trt 1

Intercept

$$Y = (\beta_0 + \underline{\beta_1}) + \beta_2 X_{2j} + \underline{\beta_3 X_{2j}} + \varepsilon_j$$

$$Y = (\beta_0 + \beta_1) + \underline{(\beta_2 + \beta_3) X_{2j}} + \varepsilon_j$$

$$\rightarrow Y = (28.198 \underline{- 46.648}) + (0.948 \underline{- 0.572}) X_2$$

Trt 2

Intercept

$$Y = (\beta_0) + \beta_2 X_{2j} + \varepsilon_j$$

$$\rightarrow Y = (28.198) + (0.948) X_2$$